

Matthew C. Ferguson

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I'm a senior machine learning engineer with an **aerospace background** and **10+ years applying machine learning, cloud systems, automation, and reliability analytics** to FAA and U.S. Air Force programs. I've cut test cycles from months to hours, improved fleet reliability, and turned **complex engineering data into business results**. I have demonstrated expertise with python, machine learning, deep learning, data analytics, and computer vision. I have a (sometimes unhealthy) obsession with learning.

EDUCATION

Virginia Polytechnic and State University, Blacksburg, VA

- ❖ **Master of Science in Computer Engineering and Machine Learning**,
Summa Cum Laude, December 2024, 4.0 GPA
- ❖ Bachelor of Science in Mechanical Engineering, *Cum Laude, December 2015*

EXPERIENCE

Senior Data Scientist, Federal Aviation Administration

July 2025, Oklahoma City, Oklahoma

- ❖ **Reduced radar correlator test cycle from months to hours** using containerization to test builds in parallel.
- ❖ Led design and development and delivery for cloud-based pipeline for radar sustainment metrics.
- ❖ Created financial automations **saving hundreds of man hours**.
- ❖ Consulting machine learning and AI SME for radar development.

Principal Data Scientist, Northrop Grumman

November 2021 – July 2025, Oklahoma City, Oklahoma

- ❖ Designed and developed reliability, availability and maintainability models of E-3 systems and components.
- ❖ Designed and developed cloud applications to host and display reliability key performance indicators.
- ❖ Performed FRACAS root causing and data analytics on maintenance actions to drive **over 60 reliability improvements** to technical orders.
- ❖ Developed unsupervised clustering machine learning model of E-3 maintenance actions.
- ❖ Developed SVM machine learning model to predict maintenance action.
- ❖ Authored major technical deliverables to communicate program outcomes to stakeholders and **win new business through RFI/RFP process**.

Reliability Data Engineer, Northrop Grumman

December 2018-November 2021, Oklahoma City, Oklahoma

- ❖ Led and developed Function, Failure, Effect, and Cause (FFEC) analysis of over 200 E-3 systems.
- ❖ Led the development of system reliability/availability models by **leveraging millions of USAF E-3 maintenance actions**.
- ❖ Led NGDS troubleshooting of USAF E-3 defects and assisted with modification installs for the E-3 AWACS.
- ❖ Led and developed reliability program plan proposal for USAF E-3G systems.
- ❖ Supported development of automation and tools for E-3 reliability and maintainability, automated production of thousands of documents using VBA and Python.

Precision Castparts Corporation Reliability Engineer

January 2017-November 2018, Wichita, KS

- ❖ Developed demand projections to **identify inventory shortages before affecting production**.

RESEARCH

- ❖ Independent Research- Developed deep learning model to predict redshift of galaxies and **achieved best ever performance** in published literature across a range of redshift values
- ❖ NASA/Turbolab – Developed and implemented efficient configurations for densely instrumented turbofan cells, developed turbofan sensors for sand ingestion
- ❖ GE Aviation – Developed embedded engine simulations used in takeoff to validate engine starts for 737 aircraft